

Workshop Overview

A complete roadmap of our Machine Learning journey.

Day 1: Foundations of Statistical ML

Module 0: Math Refresher

A friendly refresher on the mathematical concepts you'll need: Linear Algebra, Calculus, and Probability.

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Module 1: Introduction to ML

Understanding the core concepts: What is "learning"? The difference between Supervised and Unsupervised learning, and the concept of Risk.

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Module 2: Linear Regression

The "Hello World" of regression. Learn how to fit a line to data to predict continuous values like prices or temperatures.

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Module 3: k-Nearest Neighbors

A simple, intuitive algorithm for classification. "Tell me who your friends are, and I'll tell you who you are."

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Module 4: Linear Discriminant Analysis (LDA)

Find the best axis to separate two classes by maximizing distance and minimizing spread.

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Day 2: Classification & Model Health

Module 5: Logistic Regression

Don't let the name fool you—it's for classification! Learn how to predict probabilities (like "Spam" vs. "Not Spam").

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Module 6: Support Vector Machines (SVM)

Find the optimal hyperplane. Understand Margins, Support Vectors, and the Kernel Trick.

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Module 7: Overfitting (Bias vs. Variance)

The most important concept in ML. Why making a model "too smart" makes it dumb on new data.

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Module 8: Evaluation Metrics

Accuracy isn't everything. Learn about Precision, Recall, and the Confusion Matrix.

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Day 3: Unsupervised & Deep Learning

Module 9: Dimensionality Reduction (PCA)

How to simplify complex data by finding the "Principal Components" that matter most.

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Module 10: Clustering (K-Means)

Unsupervised learning. Grouping similar items together without any labels or answer keys.

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Module 11: Neural Networks

Introduction to Deep Learning. How layers of simple neurons can solve complex problems.

